

20) x | x | 1011 02

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1er Parcial de **MATEMATICA SUPERIOR**

Tema 1

1		2	3		4		5	Nota Final
1	1	1.5	1	1	0,5	1	1.5	
B	B	B	B	B	B	B	B	9 (media)

Ejercicio n° 1: a) Halle el valor de $n \in \mathbb{N} / (1 - 2j)^n = 117 - 44j$ (Justifique su forma de hallarlo)
 b) Resuelva en $\mathbb{C}$: $z^4 + 16 = 0 \wedge |z + 1 - j| \leq 1$

Ejercicio n° 2: Marque la respuesta correcta, y justifique:

$\int_0^{\infty} t \cos(3t) e^{-5t} dt =$ (Justifique utilizando Transformada de Laplace)

a) K= 4/289 ; b) 5/34 ; c) 5/1156 ; d) ∞ ; e) ninguna de las anteriores

Ejercicio n° 3: Dada $f(t) = \begin{cases} |t| & \text{si } -1 < t < 1 \\ k & \text{si } 1 < |t| < 2 \end{cases} \wedge f(t) = f(t+4)$

- a) Indique el valor de $k \in \mathbb{R}$ para que el valor medio de la señal dada sea 1. Justifique.
 b) Para $k=0$, desarrolle en Serie Trigonométrica de Fourier.

Ejercicio n° 4: Sea un sistema cuya transferencia es $G(s) = \frac{41(s^3 + s^2) + 87s + 87}{s(s+3)(s^2 + 10s + 29)}$

- a) Indique si el sistema es estable. Justifique.
 b) Grafique aproximadamente el corte de $|G(s)|$ por el eje real.
 b) Halle la respuesta del sistema a la entrada impulso $\delta(t)$

Ejercicio n° 5: Calcule la transformada Z de: $x(n) = \begin{cases} (-3)^n & \text{si } n \text{ es par} \\ 5 \cdot 2^n & \text{si } n \text{ es impar} \end{cases}$ Indicando la región de convergencia de la misma.

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1º Parcial Matemática Superior

hoja 1/3

① ② $1-2j \rightarrow [\sqrt{5}, \arctg(-2)]$



$117-44j \rightarrow [125, \arctg(-\frac{44}{117})]$

$\sqrt{5}^n = 125$

$5^{\frac{n}{2}} = 125$

$\frac{n}{2} = 3$

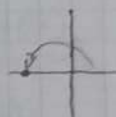
$n = 6$ ✓

$5^3 = 125$

⑥

$z^4 = -16$

$-16 \rightarrow [16; \pi]$



$\sqrt[n]{z} = \sqrt[n]{-16}$

$w_k = [\sqrt[4]{16}; \frac{\pi + 2k\pi}{4}] \quad 0 \leq k \leq 3$

$w_0 = [2; \frac{\pi}{4}] \rightarrow (\sqrt{2}, \sqrt{2})$

$w_1 = [2; \frac{3\pi}{4}] \rightarrow (-\sqrt{2}, \sqrt{2})$

$w_2 = [2; \frac{5\pi}{4}] \rightarrow (-\sqrt{2}, -\sqrt{2})$

$w_3 = [2; \frac{7\pi}{4}] \rightarrow (\sqrt{2}, -\sqrt{2})$

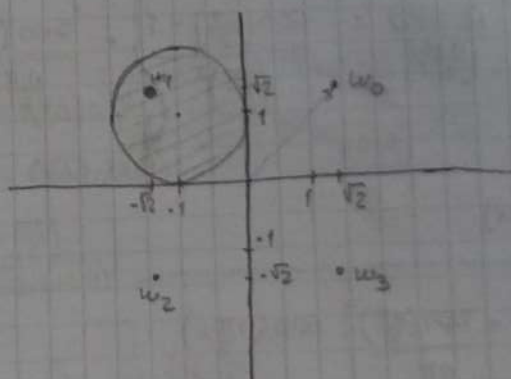
$|z+1-j| \leq 1$

$|a+bj+1-j| \leq 1$

$|a+1+(b-1)j| \leq 1$

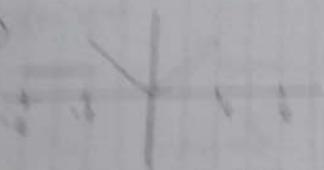
$\sqrt{(a+1)^2 + (b-1)^2} \leq 1$

$(a+1)^2 + (b-1)^2 \leq 1$ ✓



Intersección entre ambos conjuntos = $-\sqrt{2} + \sqrt{2}j$ ✓

3)



$$f(t) = f(t+4)$$

$$T = 4$$

$$\omega = \frac{\pi}{2}$$

ou par par
symétrie d'axe

$$\frac{a_0}{2} = 1$$

$$a_0 = \frac{1}{2} \int_0^4 f(t) dt = \frac{1}{2} \int_0^2 t dt + \int_2^4 2 dt = \frac{1}{2} \left[\frac{t^2}{2} \right]_0^2 + \left[2t \right]_2^4 = \frac{1}{2} \left(\frac{4}{2} \right) + 4 - 4 = \frac{1}{2} + 2 = \frac{5}{2}$$

$$\frac{a_0}{2} = 1 \Rightarrow \frac{1}{2} + 2 = 2$$

$$a_0 = 2$$

$$K = \frac{5}{2}$$

4)

$$K = 0$$



$$f(t) \Rightarrow b_n = 0$$

$$a_0 = \frac{1}{2} \int_0^4 f(t) dt = \int_0^2 t dt = \frac{t^2}{2} \Big|_0^2 = \frac{4}{2} = 2 \quad \frac{a_0}{2} = 1$$

$$a_n = \frac{1}{2} \int_0^4 f(t) \cos\left(\frac{n\pi}{2}t\right) dt = \int_0^2 t \cos\left(\frac{n\pi}{2}t\right) dt = \left[\frac{\cos\left(\frac{n\pi}{2}t\right)}{\left(\frac{n\pi}{2}\right)^2} + \frac{t \sin\left(\frac{n\pi}{2}t\right)}{\frac{n\pi}{2}} \right]_0^2 =$$

$$= \frac{\cos\left(\frac{n\pi}{2}\right)}{\left(\frac{n\pi}{2}\right)^2} + \frac{\sin\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{2}} - \frac{\cos(0)}{\left(\frac{n\pi}{2}\right)^2} = \frac{\cos\left(\frac{n\pi}{2}\right) - 1}{\left(\frac{n\pi}{2}\right)^2} + \frac{\sin\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{2}} \quad \begin{cases} -1 \\ 0 \\ 1 \end{cases}$$

$$S(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\omega x)$$

$$S(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \left[\frac{\cos\left(\frac{n\pi}{2}\right) - 1}{\left(\frac{n\pi}{2}\right)^2} + \frac{\sin\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{2}} \right] \cos\left(\frac{n\pi}{2}x\right)$$

2d) $\frac{1}{s^2 + 9}$

hoja 2/3

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$$L(\cos(3t))' = \left(\frac{s}{s^2 + 9} \right)' = \frac{s^2 + 9 - 2s^2}{(s^2 + 9)^2}$$

$$L(f(t)) = F(s)$$

$$\textcircled{2} \int_0^{\infty} \underbrace{t \cos(3t)}_{f(t)} e^{-st} dt = F(s)$$

$$F(s) = - \frac{(9 - s^2)}{(s^2 + 9)^2} = \frac{s^2 - 9}{(s^2 + 9)^2}$$

$$F(s) = \frac{s^2 - 9}{(s^2 + 9)^2} = \frac{4}{289}$$

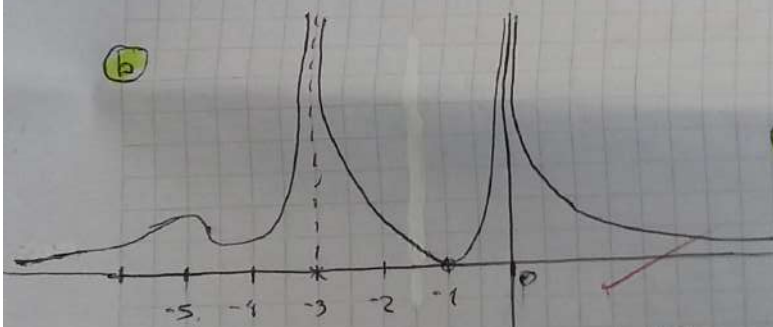
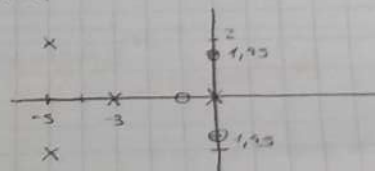
$$\textcircled{4} G(s) = \frac{41s^3 + 41s^2 + 87s + 87}{s(s+3)(s^2 + 10s + 29)} = \frac{(s+1)(41s^2 + 87)}{s(s+3)(s^2 + 10s + 29)}$$

$$\begin{aligned} z_1 &= -1 & p_1 &= 0 \\ z_2 &= \sqrt{\frac{87}{41}}j & p_2 &= -3 \\ z_3 &= -\sqrt{\frac{87}{41}}j & p_3 &= -5 + 2j \\ z_4 &= \infty & p_4 &= -5 - 2j \end{aligned}$$

$$\begin{array}{r|rrrr} 41 & 41 & 87 & 87 \\ & -41 & 0 & -87 \\ \hline -1 & 41 & 0 & 87 & 0 \end{array}$$

$$= \frac{1}{41} \frac{(s+1)(s^2 + \frac{87}{41})}{s(s+3)(s^2 + 10s + 29)}$$

$$41s^2 + 87$$



$\textcircled{2}$ estable porque todos polos estan del lado n o sobre el eje

$$\textcircled{c} x(t) = \delta(t) \Rightarrow X(s) = 1$$

$$\text{salida } y(t) \Rightarrow Y(s) = G(s) \cdot X(s)$$

$$Y(s) = \frac{1}{41} \frac{(s+1)(s^2 + \frac{87}{41})}{s(s+3)(s^2 + 10s + 29)}$$

$$Y(s) = \frac{A}{s} + \frac{B}{s+3} + \frac{Cs+D}{s^2 + 10s + 29}$$

$$41s^3 + 41s^2 + 87s + 87 = A(s+3)(s^2 + 10s + 29) + B(s^3 + 10s^2 + 29s) + C(s^3 + 3s^2) + D(s)$$

$$\begin{cases} 41 = A + B + C \\ 41 = 3A + 10B + 3C + D \\ 87 = 59A + 29B + 3D \\ 87 = 87A \end{cases}$$

$$A = 1$$

ojo con el sistema.

$$\begin{cases} B + C = 40 \\ 10B + 3C + D = 28 \\ 29B + 3D = 28 \end{cases}$$

Para no equivocarse 1908

$$\begin{aligned} 10B + 3C + D &= 29B + D \\ 10B + 20 - 3B + D &= 29B + D \\ 120 &= 22B \\ B &= \frac{60}{11} \\ C &= \frac{31}{11} \end{aligned}$$

$$B = 38$$

Sigo en hoja 3

$$D = \frac{848}{11}$$

3



$$f(z) = f(z+4)$$

$$L = \frac{1}{2}$$

5) $X(z) = (-3)^0 z^{-0} + 5 \cdot 2^1 z^{-1} + (-3)^2 z^{-2} + 5 \cdot 2^3 z^{-3}$

$$X(z) = \sum_{n=0}^{\infty} \left[\left(\frac{-3}{z} \right)^n + 5 \cdot \left(\frac{2}{z} \right)^{n+1} \right] = \sum_{n=0}^{\infty} \left(\frac{9}{z^2} \right)^n + \frac{10}{z} \left(\frac{4}{z^2} \right)^n$$

$$X(z) = \frac{1}{1 - \frac{9}{z^2}} + \frac{10}{z} \frac{1}{1 - \frac{4}{z^2}}$$

$$X(z) = \frac{z^2}{z^2 - 9} + \frac{10}{z} \frac{z^2}{z^2 - 4}$$

$$X(z) = \frac{z^2}{z^2 - 9} + 10 \frac{z}{z^2 - 4}$$

con $|z| > 3$

$$\left| \frac{9}{z^2} \right| < 1$$

$$9 < |z^2|$$

$$3 < |z|$$

mayor

$$\left| \frac{4}{z^2} \right| < 1$$

$$4 < |z^2|$$

$$2 < |z|$$



$$(4c) \quad Y(s) = \frac{1}{s} + \left(\frac{60}{s+3} + \frac{380s+848}{s^2+10s+29} \right) \cdot \frac{1}{11}$$

$$Y(s) = \frac{1}{s} + \left(\frac{60}{s+3} + \frac{380(s+5) + 848}{(s+5)^2 + 4} \right) \frac{1}{11}$$

$$Y(s) = \frac{1}{s} + \left(\frac{60}{s+3} + \frac{380(s+5-5) + 848}{(s+5)^2 + 4} \right) \frac{1}{11}$$

$$Y(s) = \frac{1}{s} + \left(\frac{60}{s+3} + \frac{380(s+5)}{(s+5)^2 + 4} - \frac{1052}{(s+5)^2 + 4} \right) \cdot \frac{1}{11}$$

$$Y(t) = 1 + \frac{60}{11} e^{-3t} + \frac{380}{11} \cos(2t) e^{-5t} - \frac{526}{11} \sin(2t) e^{-5t}$$

a corrección de cálculo